

Q5-A: Prove That a Group of Order 40 Has a Nontrivial Normal Subgroup

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Preliminaries

Definition 1 (Normal Subgroup). A subgroup $N \leq G$ is **normal** in G , written $N \trianglelefteq G$, if $gNg^{-1} = N$ for all $g \in G$. It is **nontrivial** if $N \neq \{e\}$ and $N \neq G$.

Theorem 1 (Sylow's Theorem). Let G be a finite group with $|G| = p^n m$, where p is prime and $\gcd(p, m) = 1$. Let n_p denote the number of Sylow p -subgroups of G . Then:

- (i) Sylow p -subgroups exist and have order p^n .
- (ii) Any two Sylow p -subgroups are conjugate in G .
- (iii) $n_p \equiv 1 \pmod{p}$ and $n_p \mid m$.
- (iv) A Sylow p -subgroup is normal in G if and only if $n_p = 1$.

Proof

Theorem 2. Every group G of order 40 has a nontrivial normal subgroup.

Proof. **Step 1: Factor the order and apply Sylow's Theorem.**

We have

$$|G| = 40 = 2^3 \cdot 5.$$

Apply Sylow's Theorem with $p = 5$, so that $p^n = 5$ and $m = 8$. The number of Sylow 5-subgroups satisfies:

$$n_5 \equiv 1 \pmod{5} \quad \text{and} \quad n_5 \mid 8.$$

Step 2: Determine n_5 .

The positive divisors of 8 are 1, 2, 4, 8. We check each modulo 5:

$$1 \equiv 1, \quad 2 \equiv 2, \quad 4 \equiv 4, \quad 8 \equiv 3 \pmod{5}.$$

The *only* divisor of 8 satisfying $n_5 \equiv 1 \pmod{5}$ is $n_5 = 1$. Hence:

$$n_5 = 1.$$

Step 3: Conclude normality.

Since $n_5 = 1$, there is a **unique** Sylow 5-subgroup $P_5 \leq G$. By Sylow's Theorem (iv), uniqueness implies normality:

$$P_5 \trianglelefteq G.$$

Since $|P_5| = 5$, we have $P_5 \neq \{e\}$ and $P_5 \neq G$, so P_5 is a nontrivial normal subgroup of G . $\square$

Conclusion

Every group of order 40 has a nontrivial normal subgroup.

Corollary 1. *No group of order 40 is simple.*

Proof. A simple group has no proper nontrivial normal subgroups. Since every group of order 40 contains $P_5 \trianglelefteq G$ with $|P_5| = 5$, no group of order 40 can be simple. $\square$

Remark 1. The argument is entirely elementary: among the divisors 1, 2, 4, 8 of 8, exactly one — namely 1 — is congruent to 1 modulo 5. This forces $n_5 = 1$ without any further case analysis, and the unique Sylow 5-subgroup is immediately normal.